Spin Hall - Littlewood Polynomials

1. Introduction

When studying Schur polynomials, we mentioned following identity, now referred to as a Cauchy-type identity.

$$\sum_{\substack{\lambda \\ \ell(\lambda)\leq N}} s_\lambda(u_1,\dots,u_N)\, s_\lambda(v_1,\dots,v_N) = \prod(u,v) = \prod_{i=1}^{N} (1-u_i v_i)^{-1}$$

Next, for the Hall-Littlewood polynomials

$P_\lambda(u;q) = \dfrac{1}{v_\lambda(q)} \displaystyle\sum_{\sigma\in S_n} \sigma\left(u^\lambda \prod_{i<j} \frac{u_i - q u_j}{u_i - u_j}\right)$, where $v_\lambda(q) = \displaystyle\prod_{i\geq 0} v_{m_i(\lambda)}(q)$ with

$m_i(\lambda)$ = # of parts of λ equal to i, $v_r(q) = \dfrac{(q;q)_r}{(1-t)^r}$ and $(q;q)_r = (1-q)(1-q^2)\cdots(1-q^r)$,

we also had a similar Cauchy-type identity:

$$\sum_{\substack{\lambda \\ \ell(\lambda)\leq N}} b_\lambda(q)\, P_\lambda(u;q)\, P_\lambda(v;q) = \prod_{i,j} \frac{1-q u_i v_j}{1-u_i v_j},$$

where $b_\lambda(q) = \displaystyle\prod_{i\geq 0} (q;q)_{m_i(\lambda)}$.

Finally, for Macdonald polynomials, we had

$$\sum_{\substack{\lambda \\ \ell(\lambda)\leq N}} b_\lambda(q)\, P_\lambda(u;t,q)\, P_\lambda(v;t,q) = \prod(u,v;t,q) = \prod_{i,j} \frac{(q u_i v_j;t)_\infty}{(u_i v_j;t)_\infty}$$

Our goal is to introduce a new type of symmetric function, called spin Hall-Littlewood symmetric rational function and show such functions satisfy a similar Cauchy type identity.

Def.: (Signatures) A signature of length N is a sequence of $(\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_N)$, $\lambda_i \in \mathbb{Z}$. Let Sign_N denote the set of nonnegative signatures, i.e., the ones where $\lambda_i \geqslant 0$ for all i. Let us define $\mathrm{Sign}_0 = \{\emptyset\}$.

We also define the notation $\lambda = (0^{m_0(\lambda)} 1^{m_1(\lambda)} 2^{m_2(\lambda)} \dots)$, with $m_i(\lambda) =$ # of parts in λ equal to i as before.

Define a spin Hall-Littlewood symmetric rational function by the formula

convention, $\lambda_i = 0 \Rightarrow$ product $= 1$

$$F_\lambda(u_1, u_2, \dots, u_N; q) = \sum_{\sigma \in S_N} \sigma\left(\prod_{i<j} \frac{u_i - q u_j}{u_i - u_j} \prod_{i=1}^{N} \left(\frac{1-q}{1 - s_{\lambda_i} \xi_{1+\lambda_i} u_i} \prod_{j=0}^{\lambda_i - 1} \frac{\xi_j u_i - s_j}{1 - \xi_j s_j u_i} \right) \right)$$

The action of $\sigma \in S_N$ permutes the variables u_i, but has no effect on ξ_i and s_i. The function is symmetric by definition and it is obtained as a deformation of the usual Hall-Littlewood symmetric functions. Each F_λ depends on the variables u_i, the quantum parameter q, the inhomogeneity parameters ξ_x and s_x, $x \in \mathbb{Z}_{\geq 0}$ where s_x is called the spin parameter. In general, ξ_x and s_x are thought to be arbitrary complex numbers, but for more interesting results, additional conditions will be imposed soon. Note that if $s_x = 0$ and $\xi_x = 1$ for all x, then F_λ becomes the usual Hall-Littlewood symmetric functions.

The motivation behind these functions comes from statistical mechanics, where they appear naturally as the partition functions of a specific type of integrable lattice model: higher spin six-vertex model. The definition of this model, as well as its version of the Young-Baxter Equation, will be vital to understand the Cauchy-type identity for the F_λ.

One can define a few important variations on spin Hall-Littlewood polynomials. For reasons that will be presented later, one can define the so called dual Hall-Littlewood symmetric functions in terms of the regular Hall-Littlewood functions as

$$F^{*}_{\lambda}(v_1,\dots,v_N) = \prod_{r\geq 0}\frac{(s_r^2;q)_{m_r}}{(q;q)_{m_r}}\cdot F_{\lambda}(v_1,\dots,v_N)\Big|_{\xi_x\to\xi_x^{-1}\text{ for all } x},$$

where $\lambda = (0^{m_0}1^{m_1}2^{m_2}\dots)$.

There are also the stable spin Hall-Littlewood functions, which are indexed by partitions, rather than signatures. The distinction is made as partitions only care about nonzero parts of a nondecreasing sequence of nonnegative integers, and partitions may have an arbitrary length. As a result, such stable functions may depend on an arbitrary number of variables $k\geq \ell(\lambda)$, not tied to the length of the partition. By convention, if $k<\ell(\lambda)$ the stable function is identically zero. We denote the stable functions by $\tilde{F}_{\lambda}$, $\tilde{F}^{*}_{\lambda}$.

We can define stable functions in terms of the regular functions in multiple ways. The most straightforward is (here, $\lambda=(\lambda_1\geq\lambda_2\geq\dots\geq\lambda_\ell>0$ is a partition).

$$\tilde{F}_{\lambda}(u_1,\dots,u_k) = \frac{1}{(q;q)_{k-\ell}}\cdot F_{\lambda\cup 0^{k-\ell}}(u_1,\dots,u_k)\Big|_{s_0=0}$$

$$= \sum_{\sigma\in S_k/S_\lambda}\sigma\left(\prod_{i<j}\frac{u_i-qu_j}{u_i-u_j}\prod_{i=\ell+1}^{k}\left(\frac{1-q}{1-s_0\xi_0 u_i}\right)\right) = \frac{(1-q)^{k-\ell}}{\prod_{i=\ell+1}^{N}(1-s_0\xi_0 u_i)}\cdot\left[\sum_{\sigma}\left(\prod_{i<j}\frac{u_i-qu_j}{u_i-u_j}\right)\right] = \frac{1}{\prod_i(1-s_0\xi_0 u_i)(q;q)_{k-\ell}}$$

(the bracketed sum $\to [k-\ell]_q!$)

Note that $\tilde{F}_{\lambda}$ does not depend on s_0. Similarly, dual stable functions can be defined as:

$$\tilde{F}^{*}_{\lambda}(u_1,\dots,u_k) = F^{*}_{\lambda\cup 0^{k-\ell}}(u_1,\dots,u_k)\Big|_{s_0=0}.$$

The name stable comes from the property that

$$\tilde{F}_{\lambda}(u_1,\dots,u_{k-1},0) = \tilde{F}_{\lambda}(u_1,\dots,u_{k-1}).$$

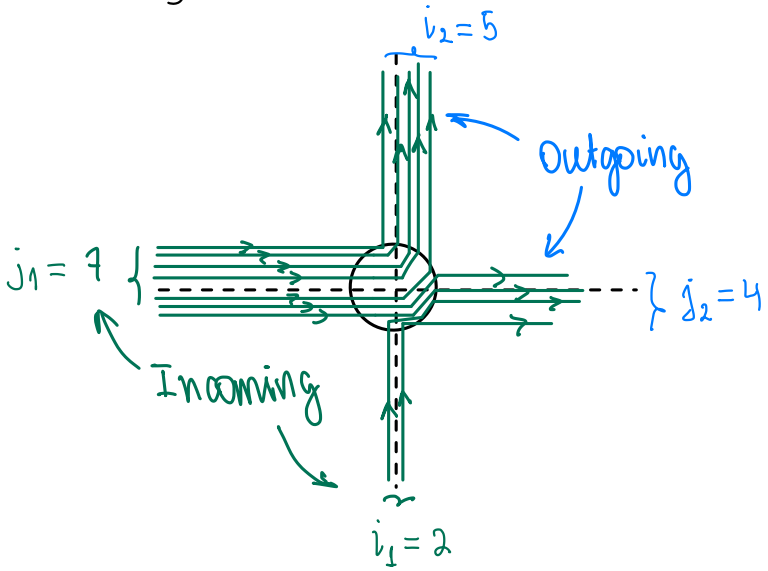
Stable spin Hall Littlewood functions satisfy a Cauchy-type identity:

$$\sum_{\lambda} \tilde{F}_{\lambda}(u_1,\dots,u_k)\ \tilde{F}^{*}_{\lambda}(v_1,\dots,v_m) = \prod_{i=1}^{n}\prod_{j=1}^{m} \frac{1-q u_i v_j}{1-u_i v_j},$$

where the sum runs over all partitions of arbitrary length.

2. The higher spin six vertex model

We admit a more general version of the usual six vertex model, where we admit each vertex can have at most I distinct incoming horizontal paths and at most J outgoing vertical paths, where $I, J \in \mathbb{Z}_{\geq 0}$. We impose the "arrow preservation property", where the number of incoming arrows equals the number of outgoing arrows. Note that this property fails at the boundaries; one way of working around this problem is to fix boundary conditions.



For the higher spin six vertex model, each vertex will be assigned a weight depending on two (complex) parameters q and s and a spectral parameter u.

Using the previous notation, the weights $w_{u,s}(i_1,i_2;j_1,j_2)$ of a given vertex must satisfy the arrow preservation rule $i_1+j_1=i_2+j_2$, i.e., so one can consider that occupied edges form up-right paths on the square grid.

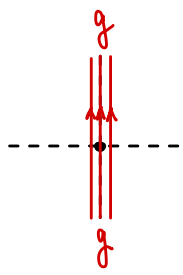
The explicit vertex weights are given by:

$$w_{u,s}(g,0;g,0) := \frac{1-sq^g u}{1-su}, \qquad w_{u,s}(g+1,0;g,1) := \frac{(1-s^2q^g)u}{1-su},$$

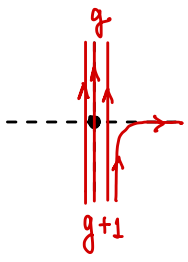
$$w_{u,s}(g,1;g,1) := \frac{u-sq^g}{1-su}, \qquad w_{u,s}(g,1;g+1,0) := \frac{1-q^{g+1}}{1-su},$$

where g is an arbitrary nonnegative integer. All other weights are assumed to be zero. Note that the weight of an empty vertex $(0,0;0,0)$ is equal to 1 and $w_{u,s}(i_1,i_2;j_1,j_2)$ is nonzero iff $j_1,j_2\in\{0,1\}$. Hence, for the higher spin six vertex model, $I=1$, while there is no limitation to J.

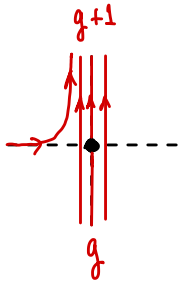
In general, the nontrivial weights fall into the following four categories:



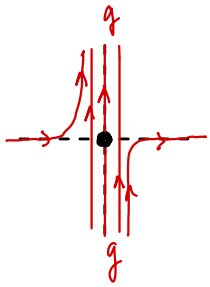
$$w_{u,s}(g,0;g,0) = \frac{1-suq^g}{1-su}$$



$$w_{u,s}(g+1,0;g,1) = \frac{(1-s^2q^g)u}{1-su}$$



$$w_{u,s}(g,1;g+1,0) = \frac{1-q^{g+1}}{1-su}$$

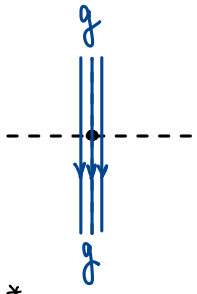


$$w_{u,s}(g,1;g,1) = \frac{u-sq^g}{1-su}$$

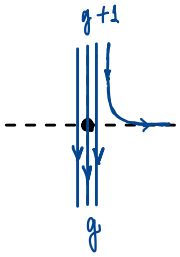
We also define dual weights $w^*_{u,v}$ for each vertex. The following formula expresses the relation between both weights.

$$W^*_{v,s}(i_1,j_1;i_2,j_2) = \frac{(s^2;q)_{i_1}(q;q)_{i_2}}{(s^2;q)_{i_2}(q;q)_{i_1}} W_{v,s}(i_2,j_1;i_1,j_2).$$

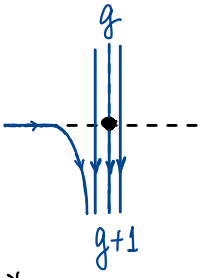
Note that the arrow preservation property now reads $i_2 + j_1 = i_1 + j_2$. Hence for the lattice with dual weights, we can think of occupied edges as forming down-right paths. The non trivial weights can be described by



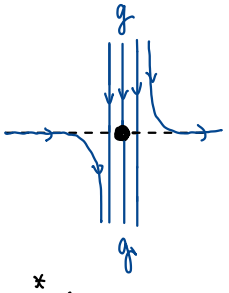
$$W^*_{v,s}(g,0;g,0) = \frac{1-svq^g}{1-sv}$$



$$W^*_{v,s}(g+1,0;g,1) = \frac{1-s^2q^g}{1-sv}$$



$$W^*_{v,s}(g,1;g+1,0) = \frac{(1-q^{g+1})v}{1-sv}$$



$$W^*_{v,s}(g,1;g,1) = \frac{v-sq^g}{1-sv}$$

Throughout this talk, we will use the following notation: the red model denotes up-right paths and the blue model denotes down-right paths.